BTS-III(R/S)-02.22-0012 Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

B

B.Tech. Degree III Semester Regular/Supplementary Examination February 2022

ME 19-205-0303 MECHANICS OF SOLIDS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

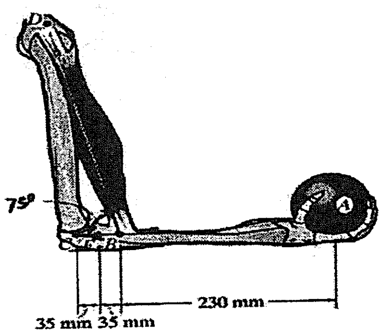
(Use of graph paper is permitted)

PART A

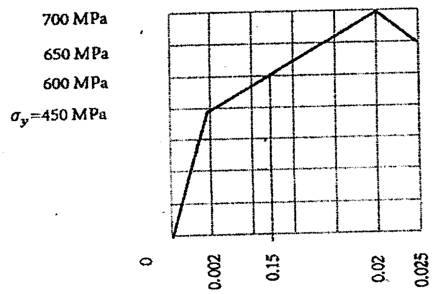
(Answer *ALL* questions)

$(8 \times 3 = 24)$

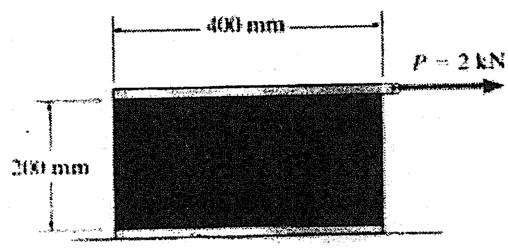
I. (a) The forearm and biceps support the 2 kg load at A. If C can be assumed as a pin support, determine the resultant internal loadings acting on the cross section of the bone of the forearm at E. The biceps pull on the bone along BD.



(b) An idealized stress-strain diagram (Not to scale) for an alloy steel specimen of 10 mm diameter and 100 mm length is shown in the figure.
Define and calculate the modulus of toughness of the material. How much energy is absorbed by the material before fracture?
If the specimen is stressed to 600 MPa and then unloaded, what is the residual strain?



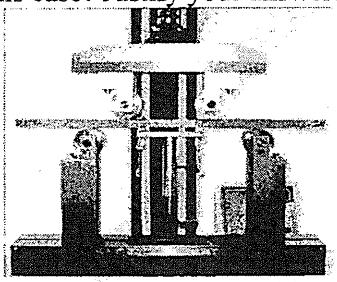
(c) An acetal polymer block is fixed to the rigid plates at its top and bottom surfaces. If the top plate displaces 2mm horizontally when it is subjected to a horizontal force P = 2 kN, determine the shear modulus of the polymer. The width of the block is 100 mm. Assume that the polymer is linearly elastic and use small angle analysis.



(P.T.O.)

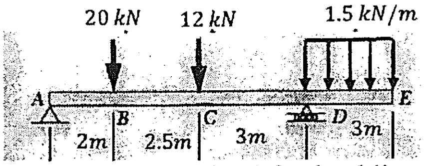
(d) A solid circular shaft of outer radius R and length L is subjected to a torque of T. Derive the expression for the maximum shear stress developed across the cross section. Clearly state the assumptions used in the derivation of the expression.

(e) What do you mean by pure bending?
The figure shows the experimental setup for flexure test of a mild steel beam of uniform cross section to find its flexural rigidity. The beam is resting on two rollers and load is applied symmetrically on the beam by moving the upper frame down. If W is the load exerted by the frame, can we apply the pure bending theory in this case? Justify your answer.



(f) Briefly explain any one theory of failure applicable to ductile materials.

(g) A beam is loaded and supported as shown. Draw the free-body diagrams of the beam at sections between AB, BC, CD, and DE and write down the expression for bending moment using Macaulay's singularity functions.



(h) Derive the expression for the critical load for buckling of column pinned at both ends.

PART B

(4 × 12 = 48)

II. (a) The following observations were recorded during the tensile testing of a specimen as per ASTM standard. (7)

Diameter of the specimen, D = 6 mm	Gauge Length, G = 30 mm
Length of reduced section, A = 36 mm	Fillet radius, R = 6 mm

The final gauge length = 45 mm
Diameter of the neck at fractured section = 4 mm

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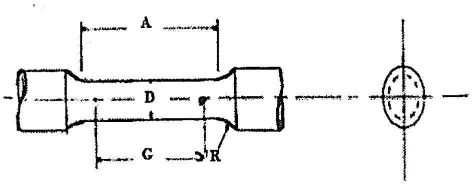


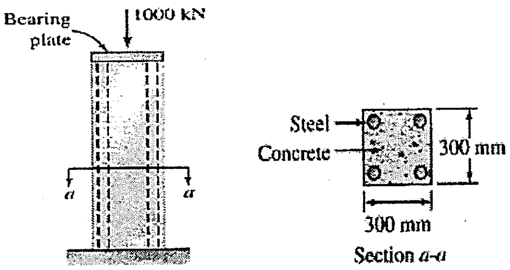
Load (kN)	0	1.5	5	9.25	11.25	11.79	11.8	12	14.25	15.25	16	16.75	17.25	17.5	16.5	15	13.25
Deflection (mm)	0	0.15	0.5	1	1.25	1.5	1.63	2	2.5	3	3.5	4.25	5	6.62	8.5	10	11

Plot the engineering stress-strain diagram and determine the following :

1. Modulus of elasticity of the material
2. Percentage elongation
3. Percentage reduction in cross-sectional area
4. Yield stress and ultimate stress.
5. The change in diameter at yield point if the Poisson's ratio, $\nu = 0.3$.

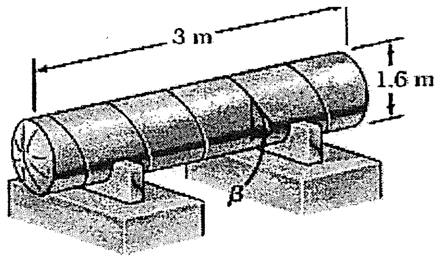
(Continued to 3)

(b) The concrete post in Figure is reinforces with four symmetrically placed steel bars, each of cross-sectional area of 900 mm^2. Compute the stress in each material when the 1000 kN axial load is applied. The moduli of elasticity are 200 GPa for steel and 14 GPa for concrete. (5)



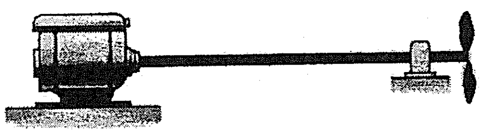
OR

III. (a) The pressure tank shown has a 8-mm wall thickness and butt-welded seams forming an angle $\beta = 20^\circ$ with a transverse plane. For a gauge pressure of 600 kPa inside the tank, determine the state of stress at a point on the outer surface of the tank. What is the normal and shear stress on planes parallel and perpendicular to the weld? What is the maximum in-plane and out-of-plane shear stress? (8)



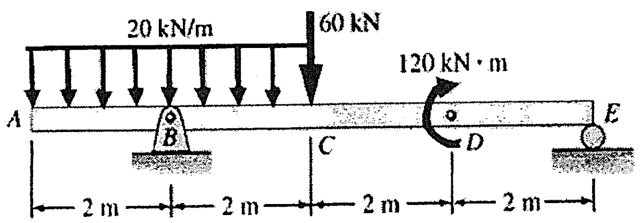
(b) Steel railroad rails 10m long are laid with end-to-end clearance of 3 mm at a temperature of 15°C. At what temperature will the rails just come in contact? Take $a = 11.7 \times 10^{-6} /^o C$ and $E = 200\ GPa$. (4)

IV. The figure shows an inboard engine, with a hollow steel drive shaft of 2.5 m length and 50 mm outer diameter. The shaft is to be designed to transmit 100 kW power at 3500 rpm. Determine the thickness of the shaft if the shear stress is not to exceed 60 MPa and the angle of twist not to exceed 5°. Take the shear modulus of steel as 83 GPa. (12)

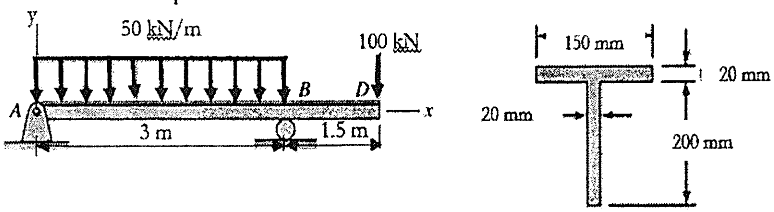


OR

V. A prismatic beam is loaded and supported as shown. Obtain the expressions for shear force and bending moment at points between AB, BC, CD, and DE. Draw the shear force and bending moment diagrams. What is the maximum bending moment developed in the beam? Is there any point of contraflexure? (12)

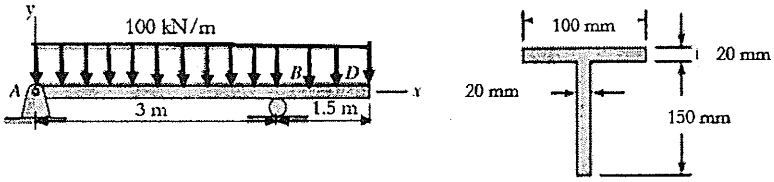


VI. Determine the maximum tensile and compressive stress developed in the beam due to the transvers loads acting on the beam as shown. Locate the cross-sectional plane from the left end where this maximum stress occurs. (12)



OR

VII. A T-beam is transversely loaded as shown. Determine the transverse shear stress developed across the cross section of the beam at a location 2m from the left support. Show the variation across the cross section. The cross-sectional dimensions of the beam are shown on the right side. (12)



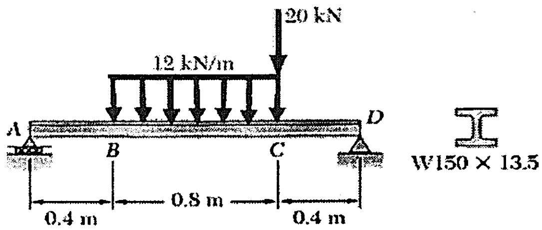
VIII. A beam is loaded and supported as shown. Determine the following. (12)

(i) Slope at A and D

(ii) Deflection at the centre and at points B and C

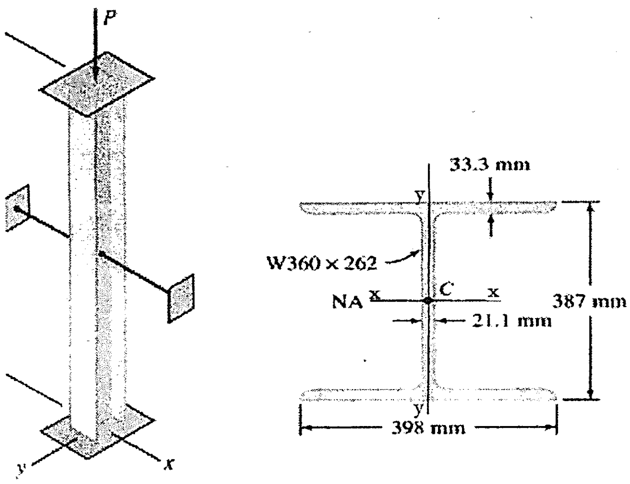
The modulus of elasticity of the material, $E = 200\ GPa$

The moment of inertia of cross section with respect to the neutral axis is $6.84 \times 10^{6}\ mm^{4}$.



OR

IX. The 7.5 m long steel column uses a W 360 × 262 section that is built-in at both ends. The dimensions of the cross section are shown on the right side. The midpoint of the column is braced by two cables that prevent displacement in the x-direction. Determine the critical value of the axial load using Euler's column formula. Use E = 200 GPa. Neglect the rounding of edges and corners. (12)



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